

SOCIOL 723: Social Statistics II

Week 13 — Causal Machine Learning

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Today

Why Naive ML Fails

Double Machine Learning

Heterogeneous Treatment Effects

Putting the Two Halves Together

- **Weeks 5–11** gave us identification: what must be true for a parameter to be causal. Almost every design ended with a nuisance function to estimate — a propensity score, an outcome regression, a first stage, a conditional trend.
- **Week 12** gave us flexible estimators for exactly such functions, which work with many covariates and unknown functional form.
- **The obvious move — plug ML into the causal estimator — fails.** Today is about why, and what fixes it.

The payoff

We get to relax the functional-form assumptions that identification arguments have been quietly leaning on all semester — AJR's linear geography control, the linear propensity score, the additive conditional trend — *without* giving up $\sqrt{n}$ inference.

The Problem

The Partially Linear Model

The workhorse specification for the rest of the course:

$$\begin{aligned} Y_i &= \theta D_i + g(W_i) + \epsilon_i, & E[\epsilon_i \mid D_i, W_i] &= 0 \\ D_i &= m(W_i) + v_i, & E[v_i \mid W_i] &= 0. \end{aligned}$$

- θ is the target: the effect of D holding W fixed, and it is assumed constant (relaxed in the last section).
- g and m are **nuisance functions** — we do not care about them, but we must handle them. They may be highly nonlinear, and W may be high-dimensional.
- Under conditional ignorability given W (Week 5), θ is the ATE.
- Week 1 solved this when g and m are linear and low-dimensional. The question now is what happens when we estimate them with lasso or a forest.

Attempt 1: Regularization Bias

The naive approach: estimate $\hat{g}$ by ML, then regress $Y_i - \hat{g}(W_i)$ on D_i .

$$\hat{\theta}_{\text{naive}} = \frac{\frac{1}{n} \sum_i D_i (Y_i - \hat{g}(W_i))}{\frac{1}{n} \sum_i D_i^2}.$$

Substituting the true model and rescaling,

$$\sqrt{n}(\hat{\theta}_{\text{naive}} - \theta) = \underbrace{\left(\frac{1}{n} \sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_i D_i \epsilon_i}_{\xrightarrow{d} \mathcal{N}(0, \Sigma)} + \underbrace{\left(\frac{1}{n} \sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_i D_i (g(W_i) - \hat{g}(W_i))}_{\text{regularization bias}}.$$

- Regularized estimators are *deliberately biased* (Week 12), and typically converge more slowly than the parametric $n^{-1/2}$.
- Then the second term does not vanish on the $\sqrt{n}$ scale — the nuisance bias is of the same order as, or larger than, the sampling noise we are trying to describe.
- **The point estimate may be fine; the inference is not.** Confidence intervals are centred in the wrong place, and collecting more data does not fix it.

Attempt 2: Partial Out Both Sides

Frisch–Waugh–Lovell says: residualize Y and D on W , then regress. With ML estimates $\hat{g}$ and $\hat{m}$,

$$\tilde{Y}_i = Y_i - \hat{\ell}(W_i), \quad \tilde{D}_i = D_i - \hat{m}(W_i), \quad \hat{\theta} = \frac{\frac{1}{n} \sum_i \tilde{D}_i \tilde{Y}_i}{\frac{1}{n} \sum_i \tilde{D}_i^2},$$

where $\ell(W) = E[Y \mid W]$.

The bias term now involves the **product** of the two estimation errors:

$$\text{bias} \approx \sqrt{n} \cdot \frac{1}{n} \sum_i (m(W_i) - \hat{m}(W_i)) (\ell(W_i) - \hat{\ell}(W_i)).$$

This vanishes when $\|\hat{m} - m\|_2 \|\hat{\ell} - \ell\|_2 = o_p(n^{-1/2})$ — for which it is sufficient that *each* converge faster than $n^{-1/4}$. Lasso and forests can do that under sparsity or smoothness assumptions. **First-order bias has become second-order.**

This Is Neyman Orthogonality Again

The residualized moment condition is

$$E[\psi(W_i; \theta, \eta)] = 0, \quad \psi = [(Y - \ell(W)) - \theta(D - m(W))] (D - m(W)),$$

and it satisfies

$$\left. \frac{\partial}{\partial \eta} E[\psi(W_i; \theta_0, \eta)] \right|_{\eta=\eta_0} = 0.$$

- You derived this in **Week 6**, for the AIPW score, using a Gateaux derivative. The partially linear case is the same statement for a different moment.
- And you derived it in **Week 1**: FWL is the linear special case, in which $\hat{m}$ is an OLS projection and the “adaptivity” of $\hat{\beta}_1$ to estimation error in the residuals is exactly orthogonality.
- **The thread of the whole course:** FWL $\rightarrow$ AIPW $\rightarrow$ DML. One idea, three levels of generality.

DML

The Second Ingredient: Cross-Fitting

Orthogonality is not quite enough. If $\hat{m}$ is fitted on the *same* observations where it is evaluated, its errors correlate with those units' outcomes — **overfitting bias**, which orthogonality does not remove.

Cross-fitting

Split the sample into K folds. For each fold k : estimate $\hat{\ell}^{(-k)}, \hat{m}^{(-k)}$ on the *other* folds, and form residuals only for observations *in* fold k . Then pool.

$$\hat{\theta}_{\text{DML}} = \frac{\frac{1}{n} \sum_k \sum_{i \in \mathcal{I}_k} (D_i - \hat{m}^{(-k)}(W_i)) (Y_i - \hat{\ell}^{(-k)}(W_i))}{\frac{1}{n} \sum_k \sum_{i \in \mathcal{I}_k} (D_i - \hat{m}^{(-k)}(W_i))^2}$$

This is the same device as cross-validation in Week 12 — but used to protect *inference*, not to select a tuning parameter.

The DML Recipe

1. Split the sample into K folds (typically $K = 5$; split on the *independent unit* if data are clustered).
2. On the training folds, estimate $\hat{\ell}(W) = \hat{E}[Y | W]$ and $\hat{m}(W) = \hat{E}[D | W]$ with any ML method — lasso, random forest, boosting, or an ensemble.
3. On the held-out fold, form residuals $\tilde{Y}_i$ and $\tilde{D}_i$.
4. Pool across folds and regress $\tilde{Y}$ on $\tilde{D}$.
5. The standard error is the usual heteroskedasticity-robust one from that final bivariate regression — **orthogonality is what makes it valid** despite the ML first stage.
6. Repeat the whole split several times with different seeds and take the median; the answer should not depend on the partition.

Chernozhukov et al. (2018). In R: DoubleML with `mlr3` learners.

What the Theory Requires — and What It Does Not

Requires

- Conditional ignorability given W (unchanged!).
- Overlap: $\hat{m}(W)$ bounded away from 0 and 1 for binary D .
- Nuisance convergence faster than $n^{-1/4}$ — e.g. approximate sparsity for lasso, smoothness for forests.
- Orthogonal moment + cross-fitting.

Does not require

- Correct functional form for g or m .
- Knowing which of the p covariates matter.
- $p \ll n$. High-dimensional W is the point.
- Consistent *model selection* — lasso may pick the wrong variables and $\hat{\theta}$ is still fine.

The assumption that does not go away

DML does not identify anything. It is an estimation technology for a parameter already identified by the arguments of Weeks 5–11. If W omits a confounder, no amount of machine learning helps.

Where DML Plugs In

The same two ingredients — an orthogonal score plus cross-fitting — apply to almost every design in this course:

Design	Nuisance functions	Week
Partially linear regression	$E[Y W], E[D W]$	1
Interactive / ATE under CIA (AIPW)	$E[Y D, W], \Pr(D = 1 W)$	6
Instrumental variables	$E[Y W], E[D W], E[Z W]$	7
Regression discontinuity	$E[Y W]$ within the bandwidth	9
Conditional DID	outcome trend and propensity, by cohort	10

- For IV: residualize Y , D , and Z on W using ML, then run 2SLS on the residuals. This is what lets us replace AJR's "linear in distance from the equator plus continent dummies" with something flexible.
- For conditional DID: this is precisely the doubly robust Callaway–Sant’Anna estimator from Week 10, with ML nuisances.

HTE

From ATE to CATE

Drop the constant-effect assumption. The target becomes the **conditional average treatment effect**

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x].$$

- Sociologically this is often the substantive question: *for whom* does college pay off, *which* neighbourhoods benefit, *who* is harmed.
- The naive approach — run subgroup regressions, report the significant ones — is a **multiple-testing disaster** and does not replicate. It is the specification search of Week 2 with more rooms to search.
- The modern approach: build an orthogonal, doubly robust *pseudo-outcome* whose conditional mean is $\tau(x)$, then use ML to estimate that conditional mean — with honest sample splitting so that the subgroups are not chosen using the same data that tests them.

The Doubly Robust Score as a Pseudo-Outcome

Take the AIPW summand from Week 6, unit by unit:

$$\Gamma_i = \hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{m}_0(X_i))}{1 - \hat{e}(X_i)}.$$

- Its *average* is the AIPW estimate of the ATE — that was Week 6.
- With the *true* nuisance functions, its conditional mean is exactly the CATE: $E[\Gamma_i \mid X_i = x] = \tau(x)$. With cross-fitted estimates it holds approximately, at a rate governed by the same product condition as before.
- So: construct Γ_i with cross-fitted nuisances, then **regress Γ_i on X_i by any method you like**.
 - Regress on a few chosen covariates → **group average treatment effects** (GATEs) with valid standard errors.
 - Regress with a forest → a nonparametric CATE surface.
 - Regress on a constant → back to the ATE.
- **This one construction unifies the whole section.**

Causal Forests

Wager and Athey (2018), Athey, Tibshirani, and Wager (2019): a random forest whose splits are chosen to maximize *heterogeneity in the treatment effect* rather than in the outcome.

- **Honesty:** within each tree, one subsample chooses the splits and a *disjoint* subsample estimates the effects in the resulting leaves. This is what delivers asymptotic normality and valid pointwise intervals — without it, leaves are selected *because* they show large effects.
- **Local centering:** residualize Y and D on X first (i.e. orthogonalize), exactly as in DML.
- **Output:** an estimate $\hat{\tau}(x)$ for every unit, with standard errors, plus variable-importance measures for *effect* heterogeneity.
- `grf::causal_forest()`.

Reporting Heterogeneity Honestly

An estimated $\hat{\tau}(x)$ that varies is not by itself evidence of heterogeneity — noise varies too.

- **Calibration test:** regress the doubly robust scores on $\hat{\tau}(X_i)$ (out of fold). A slope near 1 indicates real, well-calibrated heterogeneity; a slope near 0 says your CATE surface is noise. `grf::test_calibration()`.
- **Sorted group ATEs:** rank units by $\hat{\tau}$, form quantile groups, and report the doubly robust ATE *within each group* using held-out data. Then test whether the top and bottom groups differ.
- **Pre-specify** the moderators you care about wherever possible, and label exploratory heterogeneity as exploratory.
- **Effect heterogeneity is where causal ML is most useful and most abused.** The honest-splitting discipline is not optional.

Where This Leaves the Course

- **One idea, three appearances.** FWL (Week 1) $\rightarrow$ double robustness and Neyman orthogonality (Week 6) $\rightarrow$ DML and causal forests (today). Each generalizes the last; none replaces identification.
- **ML relaxes functional-form assumptions, not identifying assumptions.** That is a real and substantial gain — many published identification arguments lean on linearity they never defended — but it is a bounded one.
- **The binding constraint is still the design.** A causal forest on a confounded comparison estimates heterogeneity in a bias.

Next week

Synthesis, and the final project workshop. The project is due December 14.

References I

- Athey, Susan and Guido W. Imbens. 2017. “The State of Applied Econometrics: Causality and Policy Evaluation.” *Journal of Economic Perspectives* 31(2): 3–32.
- Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. “Generalized Random Forests.” *Annals of Statistics* 47(2): 1148–1178.
- Belloni, Alexandre, Victor Chernozhukov, and Christian Hansen. 2014. “High-Dimensional Methods and Inference on Structural and Treatment Effects.” *Journal of Economic Perspectives* 28(2): 29–50.
- Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. 2018. “Double/Debiased Machine Learning for Treatment and Structural Parameters.” *The Econometrics Journal* 21(1): C1–C68.
- Chernozhukov, Victor et al. 2025. *Applied Causal Inference Powered by ML and AI*. [Ch. 4 and 10]
- Wager, Stefan and Susan Athey. 2018. “Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests.” *Journal of the American Statistical Association* 113(523): 1228–1242.